

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #1

Activity Name: GPIO LED Interfacing

Assessment Rubrics:

Name of Student:		Agneay B Nair	
Registration No.:		CH.SC.U4CSE24102	
Activity Date:		18-08-2026	
Submission Date:		18-08-2026	
Component	Maximum Marks	Marks Allotted	
Program	3		
Output	4		
Viva	2		
Timely Submission	1		
Total	10		

Faculty In-charge

Write an embedded C program for ARM architecture to interface an LED. Debug and verify the program using IDE on a development computer system.

```
#include<stdio.h>
#include<LPC213x.h>

void delay(int n){
    int i;
    for(i = 0;i<n*10000;i++);
}

int main(){
    PINSEL0 = 0x00000000;
    IODIRO |= (1<<8);
    while(1){
        IOSET0 = (1<<8);
        delay(10);
        IOCLR0 = (1<<8);
        delay(10);
    }
}
```

```
1  #include<stdio.h>
2  #include<LPC213x.h>
3
4  void delay(int n){
5      int i;
6      for(i = 0;i<n*10000;i++);
7  }
8
9  int main(){
10     PINSEL0 = 0x00000000;
11     IODIR0 |= (1<<8);
12     while(1){
13         IOSET0 = (1<<8);
14         delay(10);
15         IOCLR0 = (1<<8);
16         delay(10);
17     }
18 }
```

OUTPUT:

C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 2\Code\Second.uvproj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers Disassembly

10: PINSEL0 = 0x00000000;

0x000001C0 E3A00000 MOV R0, #0x00000000

0x000001C4 E59F1044 LDR R1, [PC, #0x0044]

0x000001C8 F5810000 STR R0, [R1]

second.c

```
6   for(i = 0; i < n * 10000; i++);
7   }
8
9   int main() {
10  PINSEL0 = 0x00000000;
11  IODIR0 |= (1 << 8);
12  while(1) {
13      IOSET0 = (1 << 8);
14      delay(10);
15      IOCLR0 = (1 << 8);
16      delay(10);
17  }
18 }
```

Command

Load "C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 2\Code\

*** Restricted Version with 32768 Byte Code Size Limit

*** Currently used: 552 Bytes (1%)

Call Stack + Locals

Name	Locati...	Type
n	0x0000...	int f0

ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess

Call Stack + Locals Memory 1

Real-Time Agent: Target Stopped Simulation t1: 0.00007217 sec L:10 C:1 CAP NUM SCRL OVR R/W

